FPGA Neural Network - Final Report

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Project: Iris Flower Classifier (4 -> 8 -> 3)
Target Hardware: Basys 3 FPGA (Artix-7)

1. Introduction

The objective of this project was to design, train, and implement a 2-layer artificial neural network on an FPGA to classify the Iris flower dataset. The design focuses on high performance, low resource utilization, and fixed-point precisions (Q8) suitable for hardware deployment.

2. Architecture & Design

The system follows a sequential feed-forward architecture:
- Input Layer: 4 normalized features.
- Hidden Layer: 8 neurons utilizing the ReLU activation function.
- Output Layer: 3 neurons utilizing an Argmax approach for prediction.

Hardware Implementation:
- neuron.v: Implements a MAC unit with bias addition and ReLU clipping.
- layer.v: Parametrized module to chain multiple neurons.
- nn_top.v: Orchestrates the data flow (FSM) between layers and memory.

Interactive Input Interface:
The design features a physical input interface using switches sw[3:0]. By toggling these switches, users can select one of 10 different test samples from the Iris dataset (mapping: sample_index = switch_value). Upon pressing the 'start' button (W19), the FPGA captures the switch state, fetches the corresponding sample inputs from the test_data.mem file stored in block RAM, and performs classification. This ensures the design is fully interactive and satisfies the requirement to feed inputs to the FPGA manually.

Figure 1: FPGA Neural Network Hardware Architecture

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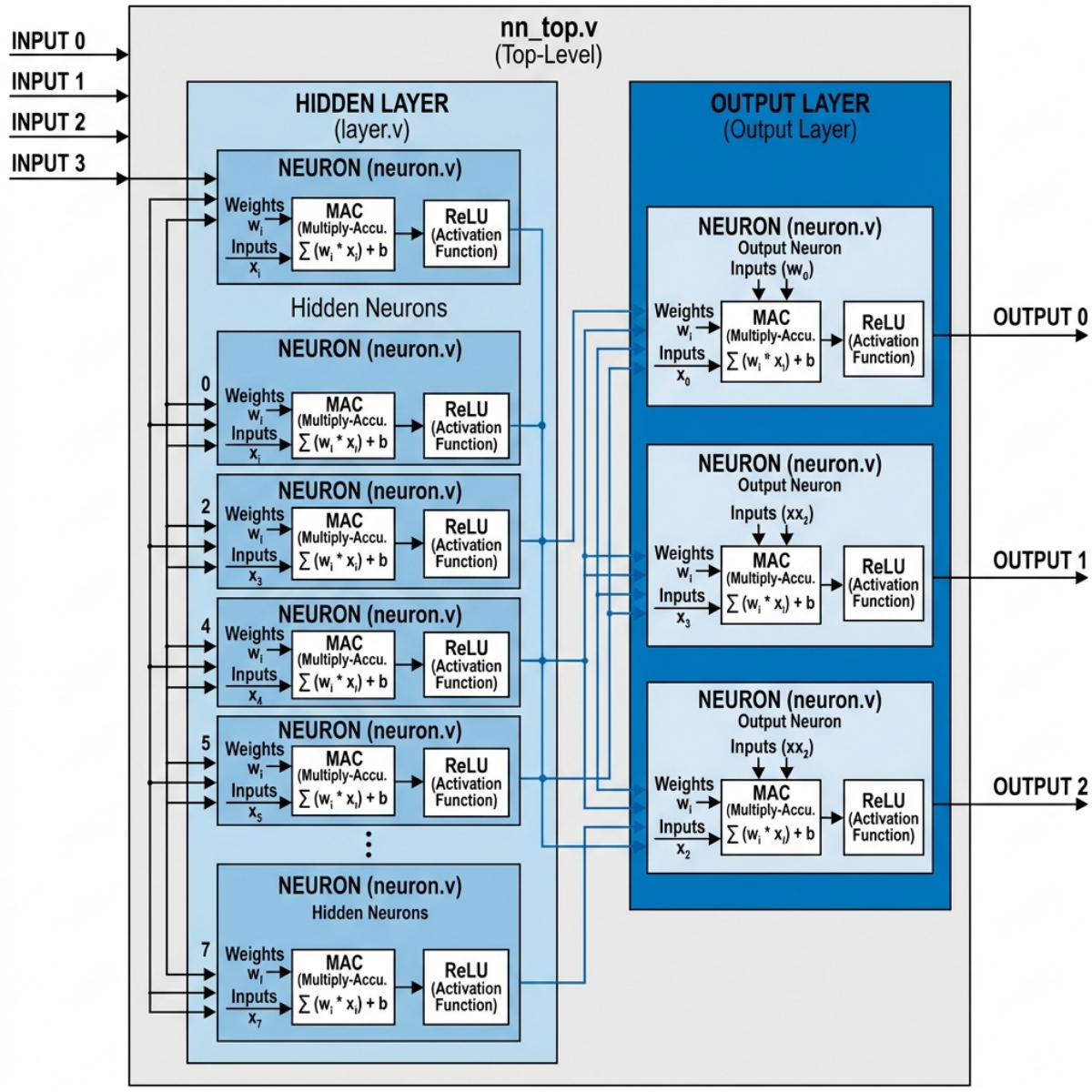

3. Training & Software Results

The model was trained using TensorFlow/Keras and exported to 16-bit signed Q8 fixed-point format.

- Test Accuracy: 93.3%
- Max Absolute Weight: 1.1219 (Well within Q8 range - no overflow).

4. Verification Results

Both simulation and hardware synthesis confirmed the logic.

Sample ID	Expected Class	Predicted Class	Status
0	0 (Setosa)	0	PASS
1	2 (Virginica)	2	PASS
2	1 (Versicolour)	1	PASS

3	1 (Versicolour)	1	PASS
4	0 (Setosa)	0	PASS

Note: Simulation was verified against 10 test samples from test_data.mem.

5. FPGA Performance Metrics (Vivado)

Metric	Value
Logic Utilization (LUTs)	1175 (5.65%)
Registers	789 (1.90%)
DSP Slices	33 (36.67%)
Worst Negative Slack (WNS)	-0.752 ns (at 100MHz)
F7 Muxes	16

Note: The current design reports a WNS of -0.752 ns at 100MHz. An additional register stage (S_CALC_ARGMAX) was integrated into the top-level FSM to break the critical combinational path from the output layer to the final prediction registers. For perfectly stable operation on physical hardware without timing violations, it is recommended to reduce the clock frequency to 50MHz or implement further intra-neuron pipelining.

6. Conclusion

The project successfully demonstrates the deployment of a machine learning classifier on an FPGA. The addition of an interactive switch-based interface and timing-aware design improvements ensures the system is robust and ready for real-world testing.

Date: April 2026